

Portable Power Monitoring Device

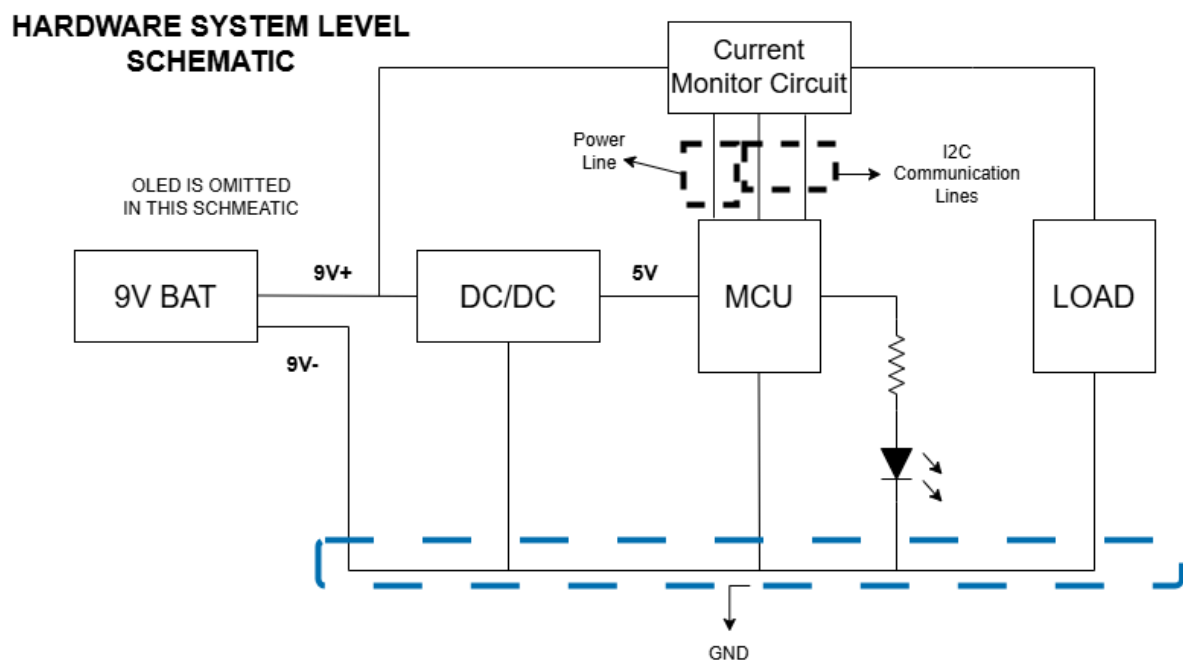
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Background/Motivation

The continued growth of modern computing systems has driven a rising demand for efficient, intelligent, and reliable power electronics. This portable power monitoring device is intended to provide hands-on insight into how real-world smart power systems measure, monitor, and respond to electrical conditions in practical applications.

Hardware Breakdown

Abstracted Schematic

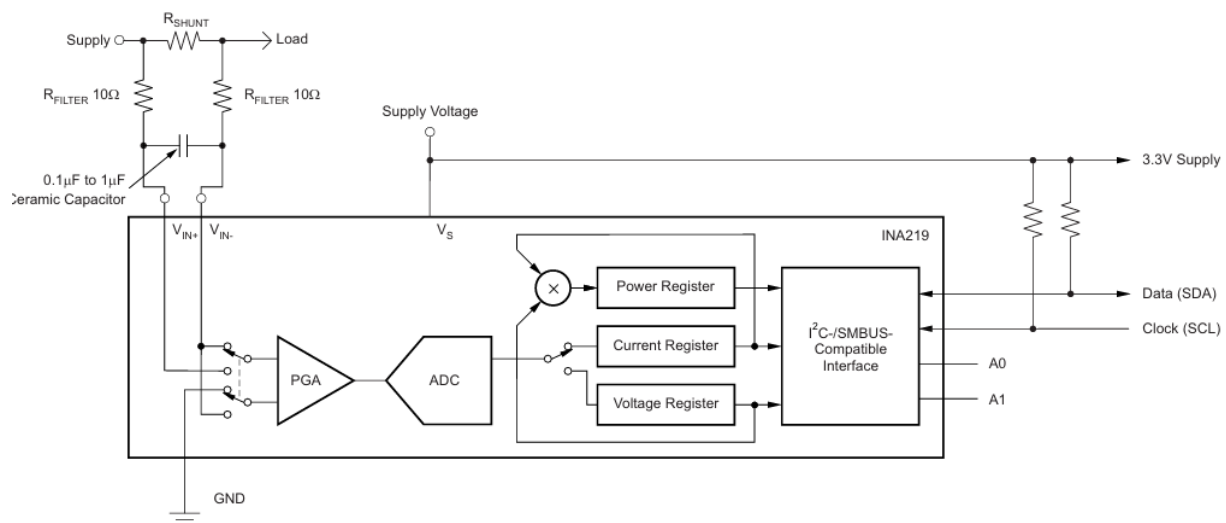


Explanation

Since this device is intended to be a demonstration of a generic power monitoring device, for safety reasons, the demonstration uses only a 9V DC battery as the power supply. The limitation of the scope for this project is taken into consideration, and in later iterations of designing this device, it is considered to add a potential protection circuit at the front-end to be able to handle higher voltages.

Microcontroller (MCU) of choice is an ESP32, it could be powered by either a 3.3V dc or a 5V dc supply, for the case of our application, we chose to use an DC-to-DC convertor to step down the supplying 9V into 5V to supply the microcontroller. The ESP32 handles communications with the current monitor circuit, as well as the online display of monitored data, on the portable device itself, an OLED screen is used to show real time data, while an LED as a demonstration if overcurrent event is captured.

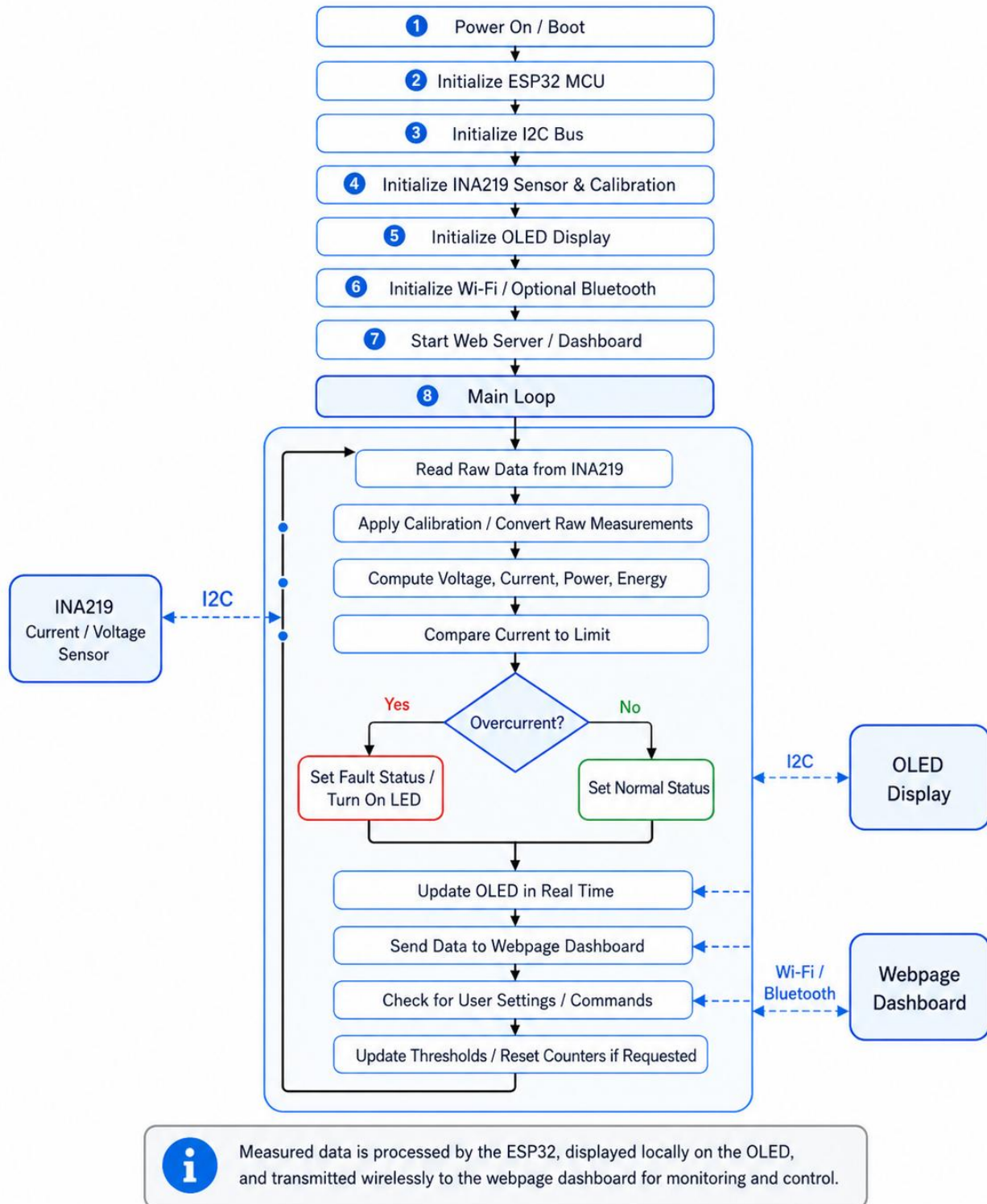
The current monitor circuit shown in the schematic is mainly handled by INA219. It is a digital current sense amplifier with an I2C communication interface. We are putting together a PCB according to the schematic in the data sheet. However, with the time constraint, there is a concern that we won't be able to get the physical board in-time at the date of showcase. So, we found breakout board option online which included input filter across the shunt resistor at the differential voltage inputs.



The load used in our demonstration is a 2-position switch linked with a 470 ohms resistor in one rail, and a 50 ohms resistor in the other rail. The 470 ohms resistor rail is intended to demonstrate the normal operating mode, whereas the 50 ohms resistor rail is intended to demonstrate the overcurrent scenario. In future iterations, a p-channel MOSFET-based driver circuit will be implemented between the current monitoring circuit and the load to cut the connections between power supply and the load if overcurrent happens.

Software Breakdown

System-Level Flowchart



Key Features

1. Real-Time Current/Power Monitor

By mapping the INA219 with ESP32 through I2C communication interface correctly, real-time current data would be captured. In INA219 datasheet, it is stated that power measurement is directly available as well. Since load used in this project has known resistance value, power calculation is also available in this design. A way for us to validate data quality is to use the measured power data from the INA219 sensor and the measured current data, to investigate if $Power = Resistance * Current^2$

2. Local OLED Display

Real-time current and power data can be read from an OLED display.

3. Webpage Dashboard

- Shows how current and power data change through time.
- Update Current Threshold / Reset Timer Counter Available.
- Overcurrent Alarm
- Data Exportation as CSV file

Team Contributions

Nomi:

- Hardware Design and Implementation
- 3D Print Enclosures for the Device and for the Dummy Load
- Sensor Calibration and Overcurrent Logic for Firmware Development

Ephram:

- Developing the Webpage Dashboard
- Mapping ESP32 with the Webpage Dashboard
- Handling I2C communication between Sensor and OLED, and Data Processing